Mini Project Report

Air Quality Index (AQI) Monitoring

Submitted By

Name: A. jithendra

Reg No: 11249A018

CSE Department | Section S5

R-Programming Lab

SCSVMV UNIVERSITY

Experiment 3: Air Quality Index (AQI) Monitoring

Aim

To collect, analyze, and visualize Air Quality Index (AQI) data from multiple cities using R programming, and to interpret pollution levels using bar charts, histograms, box plots, and pie charts.

Scenario / Background

- Environmental monitoring agencies collect daily AQI readings from sensors placed across major Indian cities.
- The AQI value reflects the concentration of key pollutants: PM2.5, PM10, CO, NO2, SO2, and O3.
- Government bodies and the public use AQI categories (Good, Moderate, Unhealthy, Very Unhealthy, Hazardous) to decide health advisories.
- Analyzing AQI trends helps identify cities with chronic pollution problems and the most common primary pollutants.
- Visualizing the data allows policymakers to take targeted action to improve air quality and public health outcomes.

Sample Dataset

The dataset contains 10 records from 5 Indian cities collected over 10 consecutive days.

Station ID	Date	City	AQI Value	Pollutant Conc. (µg/m³)	Primary Pollutant	AQI Category
S001	2024-01-15	Chennai	142	45	PM2.5	Unhealthy
S002	2024-01-16	Mumbai	89	30	PM10	Moderate
S003	2024-01-17	Delhi	210	60	PM2.5	Very Unhealthy
S004	2024-01-18	Kolkata	55	20	CO	Good
S005	2024-01-19	Bangalore	78	25	NO2	Moderate
S006	2024-01-20	Chennai	165	50	PM2.5	Unhealthy
S007	2024-01-21	Mumbai	38	10	SO2	Good
S008	2024-01-22	Delhi	295	80	PM2.5	Hazardous
S009	2024-01-23	Kolkata	110	35	PM10	Unhealthy for Sensitive
S010	2024-01-24	Bangalore	62	22	O3	Moderate

R Program (with line-by-line comments)

```
# ============================================
# Experiment 3: Air Quality Index Monitoring
# ============================================

# Step 1: Load required libraries
library(ggplot2)   # For creating visualizations
library(dplyr)     # For data manipulation

# Step 2: Define the AQI dataset inline
aqi_data <- data.frame(
  StationID = c("S001","S002","S003","S004","S005",
"S006","S007","S008","S009","S010"),
  Date      = as.Date(c("2024-01-15","2024-01-16","2024-01-17",
"2024-01-18","2024-01-19","2024-01-20",
"2024-01-21","2024-01-22","2024-01-23",
"2024-01-24")),
  City      = c("Chennai","Mumbai","Delhi","Kolkata","Bangalore",
"Chennai","Mumbai","Delhi","Kolkata","Bangalore"),
  AQI       = c(142, 89, 210, 55, 78, 165, 38, 295, 110, 62),
  Pollutant_Conc = c(45, 30, 60, 20, 25, 50, 10, 80, 35, 22),
  Primary_Pollutant = c("PM2.5","PM10","PM2.5","CO","NO2",
"PM2.5","SO2","PM2.5","PM10","O3"),
  Category  = c("Unhealthy","Moderate","Very Unhealthy","Good",
"Moderate","Unhealthy","Good","Hazardous",
"Unhealthy for Sensitive","Moderate")
)

# Step 3: View the dataset
View(aqi_data)
cat("AQI Dataset Loaded Successfully\n")
print(summary(aqi_data))

# Step 4: Bar Chart - Average AQI by City
avg_aqi <- aqi_data %>%
  group_by(City) %>%
  summarise(Avg_AQI = mean(AQI))   # Compute mean AQI per city

ggplot(avg_aqi, aes(x = City, y = Avg_AQI, fill = City)) +
  geom_bar(stat = "identity") +    # Draw bars with actual values
  labs(title = "Average AQI by City",
       x = "City", y = "Average AQI") +
  theme_minimal() +
  theme(legend.position = "none")  # Remove legend (redundant with x-axis)

# Step 5: Histogram - AQI Value Distribution
ggplot(aqi_data, aes(x = AQI)) +
  geom_histogram(binwidth = 40, fill = "steelblue", color = "black") +
  labs(title = "Distribution of AQI Values",
       x = "AQI", y = "Frequency") +
  theme_minimal()

# Step 6: Box Plot - AQI spread per City
ggplot(aqi_data, aes(x = City, y = AQI, fill = City)) +
  geom_boxplot() +                 # Show median, IQR, and outliers
  labs(title = "AQI Box Plot by City",
       x = "City", y = "AQI") +
  theme_minimal() +
  theme(legend.position = "none")
```

```
# Step 7: Pie Chart - Proportion of AQI Categories
category_counts <- table(aqi_data$Category)   # Count each category
pie(category_counts,
    main = "AQI Category Distribution",
    col  = rainbow(length(category_counts)))  # Assign distinct colors
```

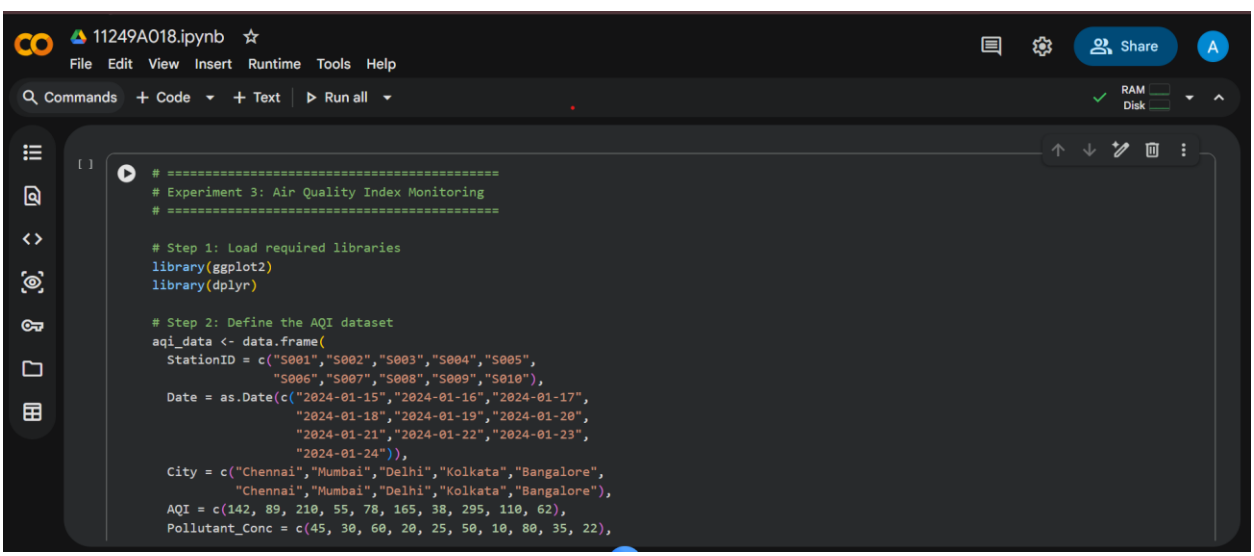
Output / Observation

- Bar Chart: Delhi records the highest average AQI (~252), followed by Chennai (~153), indicating severe pollution levels. Bangalore and Kolkata show relatively lower average AQI values.
- Histogram: The AQI distribution is right-skewed, with most readings clustered between 38 and 165. One extreme outlier (AQI = 295 for Delhi) highlights hazardous episodes.
- Box Plot: Delhi shows the widest interquartile range and highest median AQI, confirming persistent air quality issues. Mumbai and Bangalore have compact, lower-range distributions.
- Pie Chart: PM2.5 is the dominant primary pollutant (~40% of records), followed by PM10 and other pollutants, reflecting the major role of particulate matter in urban air pollution.
- Overall Insight: The analysis reveals that metro cities like Delhi face chronic high-AQI conditions, primarily driven by PM2.5. Smaller cities like Bangalore maintain relatively better air quality. Monitoring such trends is critical for data-driven environmental policymaking.

Graphs Representation

The following graphs are generated by executing the above R program:

- Graph 1: Bar Chart – Average AQI by City
- Graph 2: Histogram – Distribution of AQI Values
- Graph 3: Box Plot – AQI Spread per City
- Graph 4: Pie Chart – AQI Category Distribution



11249A018.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
  Pollutant_Conc = c(45, 30, 60, 20, 25, 50, 10, 80, 35, 22),
  Primary_Pollutant = c("PM2.5","PM10","PM2.5","CO","NO2",
                        "PM2.5","SO2","PM2.5","PM10","O3"),
  Category = c("Unhealthy","Moderate","Very Unhealthy","Good",
               "Moderate","Unhealthy","Good","Hazardous",
               "Unhealthy for Sensitive Groups","Moderate")
)

# Step 3: View dataset
print("AQI Dataset Loaded Successfully")
print(summary(aqi_data))

# Step 4: Average AQI by City (sorted)
avg_aqi <- aqi_data %>%
  group_by(City) %>%
  summarise(Avg_AQI = mean(AQI)) %>%
  arrange(desc(Avg_AQI))

# Reorder factor levels
avg_aqi$City <- factor(avg_aqi$City, levels = avg_aqi$City)

# Bar Chart
```

11249A018.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
# Bar Chart
ggplot(avg_aqi, aes(x = City, y = Avg_AQI, fill = City)) +
  geom_bar(stat = "identity") +
  labs(title = "Average AQI by City",
       x = "City", y = "Average AQI") +
  theme_minimal() +
  theme(legend.position = "none")

# Step 5: Histogram - AQI Distribution
ggplot(aqi_data, aes(x = AQI)) +
  geom_histogram(fill = "steelblue", color = "black", bins = 5) +
  labs(title = "Distribution of AQI Values",
       x = "AQI", y = "Frequency") +
  theme_minimal()

# Step 6: Box Plot - AQI by City
ggplot(aqi_data, aes(x = City, y = AQI, fill = City)) +
  geom_boxplot() +
  labs(title = "AQI Box Plot by City",
       x = "City", y = "AQI") +
  theme_minimal() +
  theme(legend.position = "none")
```

11249A018.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
       x = "City", y = "AQI") +
  theme_minimal() +
  theme(legend.position = "none")

# Step 7: Pie Chart (Improved using ggplot)
category_counts <- table(aqi_data$Category)
category_df <- as.data.frame(category_counts)
colnames(category_df) <- c("Category", "Count")

ggplot(category_df, aes(x = "", y = Count, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "AQI Category Distribution") +
  theme_void()

# Step 8: AQI Trend Over Time
ggplot(aqi_data, aes(x = Date, y = AQI, color = City, group = City)) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  labs(title = "AQI Trend Over Time",
       x = "Date", y = "AQI") +
  theme_minimal()
```

11249A018.ipynb

File Edit View Insert Runtime Tools Help

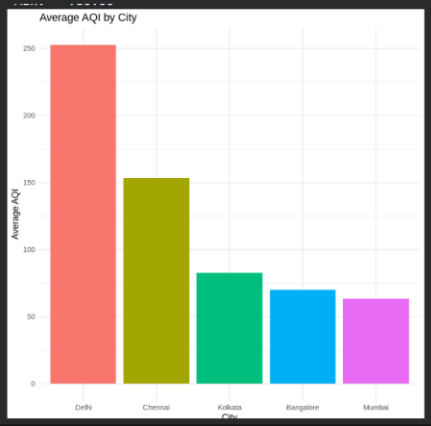
Commands + Code + Text Run all

```
[1] "AQI Dataset Loaded Successfully"
  StationID             Date               City               AQI
 Length:10          Min.   :2024-01-15   Length:10          Min.   : 38.0
 Class :character   1st Qu.:2024-01-17   Class :character   1st Qu.: 66.0
 Mode  :character   Median :2024-01-19   Mode  :character   Median : 99.5
                    Mean   :2024-01-19                      Mean   :124.4
                    3rd Qu.:2024-01-21                      3rd Qu.:159.2
                    Max.   :2024-01-24                      Max.   :295.0
 Pollutant_Conc  Primary_Pollutant    Category
 Min.   :10.00   Length:10          Length:10
 1st Qu.:22.75   Class :character   Class :character
 Median :32.50   Mode  :character   Mode  :character
 Mean   :37.70
 3rd Qu.:48.75
```

11249A018.ipynb

File Edit View Insert Runtime Tools Help

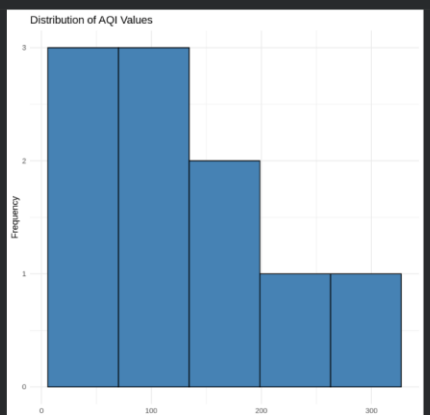
Commands + Code + Text Run all

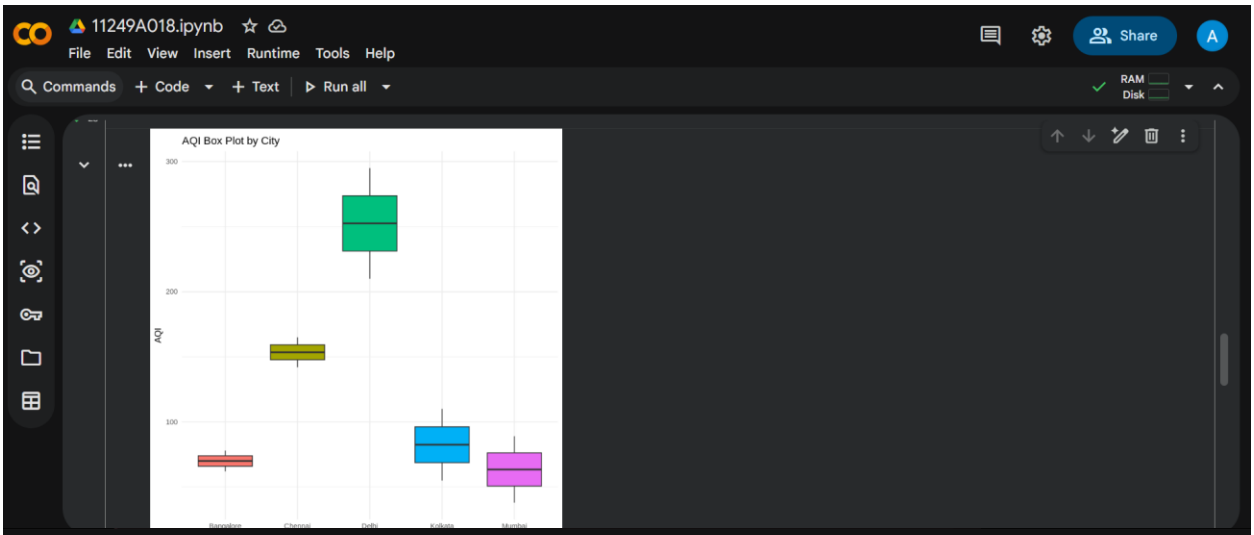


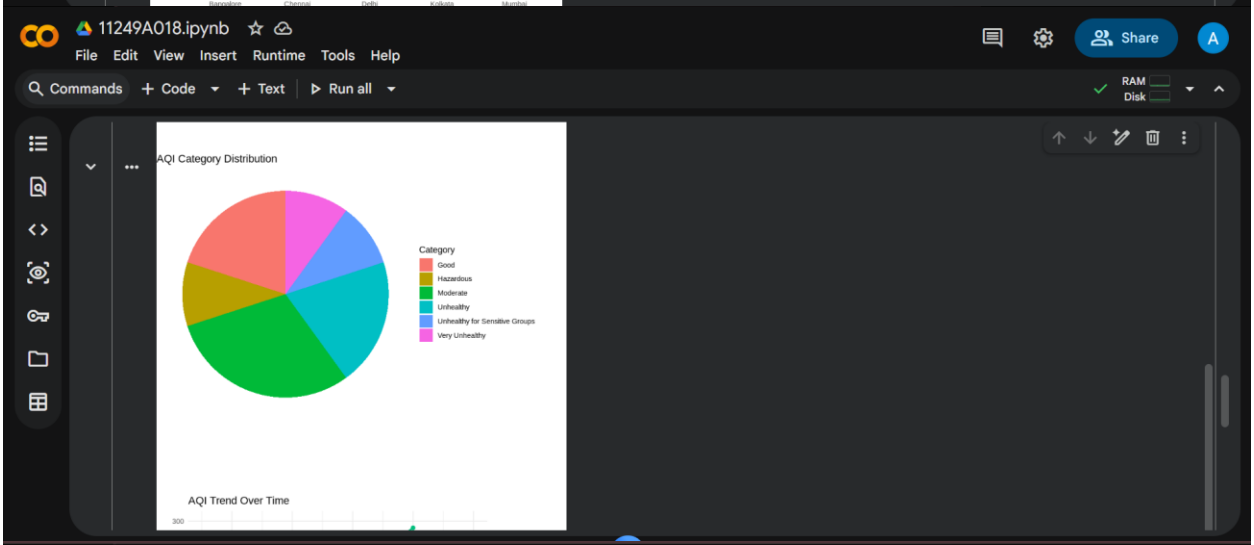
11249A018.ipynb

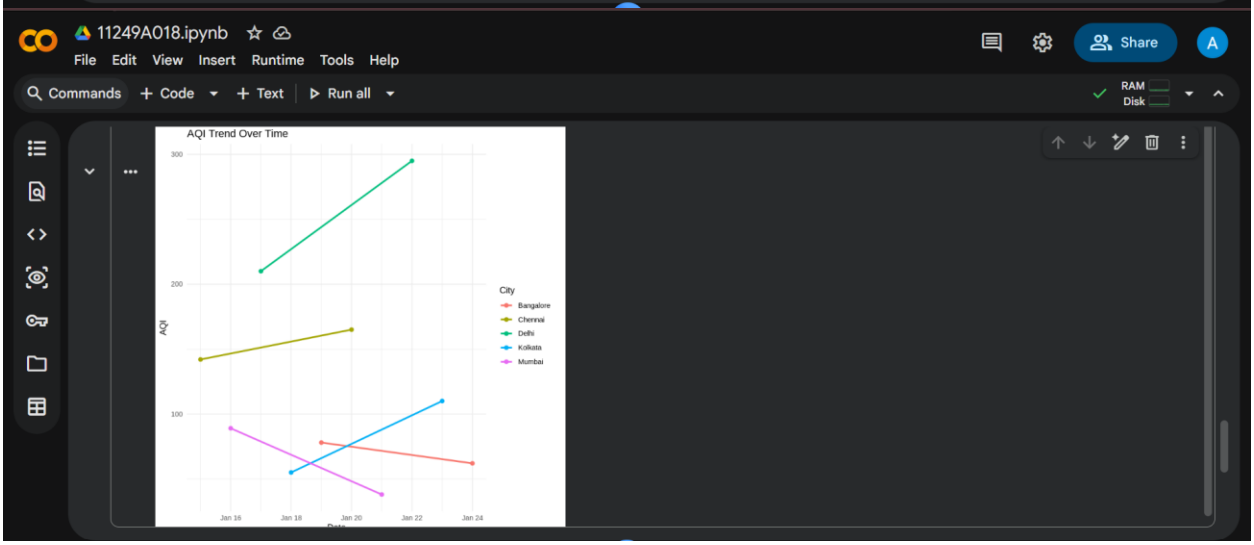
File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all




11249A018.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
RAM
Disk
AQI Box Plot by City
300
200
100
AQI
Bangalore
Chennai
Delhi
Kolkata
Mumbai


11249A018.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
RAM
Disk
AQI Category Distribution
Category
Good
Hazardous
Moderate
Unhealthy
Unhealthy for Sensitive Groups
Very Unhealthy
AQI Trend Over Time
300


11249A018.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
RAM
Disk
AQI Trend Over Time
300
200
100
AQI
Jan 16
Jan 18
Jan 20
Jan 22
Jan 24
City
Bangalore
Chennai
Delhi
Kolkata
Mumbai

Submitted By

Name : A.jithendra

Reg No: 11249A018

CSE, II Year | Section S5

R-Programming Lab

SCSVMV UNIVERSITY